Machine Code Recognition

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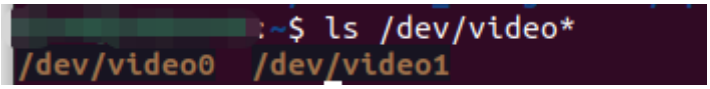
Hardware Connection

Plug the USB camera into the motherboard's USB port. Open the system terminal and enter the following command to check the camera's device number. If /dev/video0 appears, it's normal.

```
ls /dev/video*
```



Install Driver Libraries

```
sudo apt install libzbar-dev
sudo pip3 install pyzbar
sudo pip3 install pupil-apriltags
```

Example Code

```
from pupil_apriltags import Detector
import cv2 as cv
from pyzbar.pyzbar import decode
import time
import numpy as np
from PIL import Image, ImageDraw, ImageFont

# 创建检测器实例，可指定参数
# Create a detector instance, specifying parameters.
at_detector = Detector(
    families="tag36h11",  # Specify tag family
    nthreads=1,
    quad_decimate=1.0,
    quad_sigma=0.0,
    refine_edges=1,
    decode_sharpening=0.25,
    debug=0
)


def decodeDisplay(image):
    gray = cv.cvtColor(image, cv.COLOR_BGR2GRAY)
    # Perform detection
```

```
    results = at_detector.detect(gray)
    for tag in results:
        # 提取机器码的边界框的位置
        # Extract the position of the boundary box of the apriltag
        (c1, c2, c3, c4) = tag.corners
        x = int(c4[0])
        y = int(c4[1])
        x1 = int(c2[0])
        y1 = int(c2[1])
        cv.rectangle(image, (x, y), (x1, y1), (225, 0, 0), 5)

        # 绘出图像上数据和类型
        # Draw the data and type on the image
        pilimg = Image.fromarray(image)
        # 创建画笔
        # create brush
        draw = ImageDraw.Draw(pilimg)
        # # 参数1：打印坐标，参数2：文本，参数3：字体颜色
        # Parameter 1: print coordinates, parameter 2: text, parameter 3: font
color
        text = str(tag.tag_family).lstrip('b\'').rstrip('\'') + ' - ' +
str(tag.tag_id)
        draw.text((x, y - 40), text, fill=(255, 0, 0))

        # 绘出标签码的中心点
        # Draw the center of the apriltag
        tag_center = tag.center
        radius = 5
        draw.ellipse([(tag_center[0]-radius, tag_center[1]-radius),
(tag_center[0]+radius, tag_center[1]+radius)], fill="red")

        # # PIL图片转cv2 图片
        # PIL picture to CV2 picture
        image = cv.cvtColor(np.array(pilimg), cv.COLOR_RGB2BGR)
        # 向终端打印信息
        # Print information to terminal
        print(f"Tag Family: {tag.tag_family}")
        print(f"Tag ID: {tag.tag_id}")
        print(f"Center: {tag.center}")
        print(f"Corners: {tag.corners}")
        print("---")
    return image

if __name__ == '__main__':
    capture = cv.VideoCapture(0)
    cv_edition = cv.__version__
    if cv_edition[0] == '3': capture.set(cv.CAP_PROP_FOURCC,
cv.VideoWriter_fourcc(*'XVID'))
    else: capture.set(cv.CAP_PROP_FOURCC, cv.VideoWriter.fourcc('M', 'J', 'P',
'G'))
    capture.set(cv.CAP_PROP_FRAME_WIDTH, 640)
    capture.set(cv.CAP_PROP_FRAME_HEIGHT, 480)
    print ("capture get FPS : ",capture.get(cv.CAP_PROP_FPS))
    while capture.isOpened():
        start = time.time()
        ret, frame = capture.read()
        action = cv.waitKey(10) & 0xFF
        frame = decodeDisplay(frame)
```

```
        end = time.time()
        fps = 1 / (end - start)
        text = "FPS : " + str(int(fps))
        cv.putText(frame, text, (30, 30), cv.FONT_HERSHEY_SIMPLEX, 0.6, (100,
200, 200), 1)
        cv.imshow('frame', frame)
        if action == ord('q') or action == 113: break
    capture.release()
    cv.destroyAllWindows()
```

Running the Program

Open your desktop terminal and run the program:

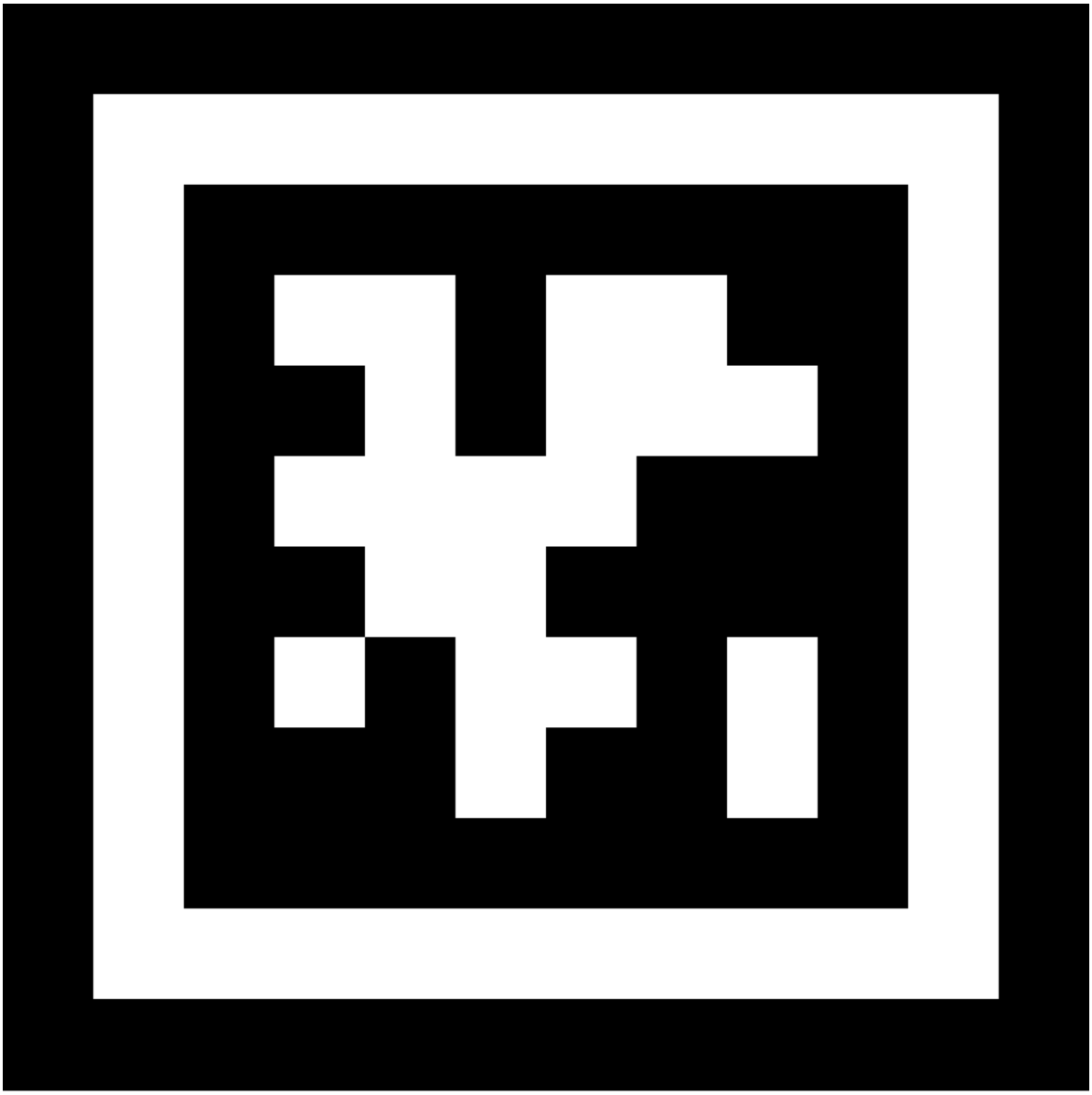
```
~/content_recognition/apriltag/
python apriltag_parsing_usb.py
```

Note: Because the program requires a display screen, do not run it in a remote SSH terminal. Please use the native desktop or VNC remote desktop to run the program.

Program Effect

After the program runs, it automatically opens the USB camera. Point the camera at the Apriltag machine code, and the machine code will be highlighted and printed in the terminal.

Example Machine Code Image:



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